

INDIAN SPACE OLYMPIAD

2026 MOCK QUESTION PAPER

JUNIOR LEVEL

CLASSES V-VI

5 Mock Papers | 50 Questions Each | 250 Questions

Strictly aligned to the ISO Junior Level syllabus for Classes V-VI. Questions 41-50 in every paper

Indian Space School

How to Use These Mock Papers

These five papers are written specifically for students of Classes V and VI. The questions remain within the supplied Junior Level, while the final ten questions of each paper provide a stronger Olympiad challenge.

Each paper contains 50 single-correct multiple-choice questions. Questions 1-40 test core knowledge, observation and application, while the final ten questions provide a stronger Olympiad challenge.

Suggested time: 60 minutes per paper. Suggested marking: 1 mark for every correct answer and no negative marking during the examination. Students should attempt the questions independently before checking the answer key.

Syllabus Coverage

Unit	Coverage
Fundamentals of Astronomy	night sky, directions, constellations, celestial sphere, telescopes, observations and events
The Solar System	Solar System formation, Sun, planets, dwarf planets, moons, asteroids, comets and meteors
Stars, Galaxies and the Universe	star properties, constellations, clusters, nebulae, galaxies and the Universe
Earth and Moon	Earth structure and atmosphere, rotation, revolution, seasons, time zones, Moon phases, eclipses and tides
Satellites and Space Technology	natural and artificial satellites, communication, navigation, weather, remote sensing and applications
Rockets and Launch Vehicles	rocket principles, stages, payloads, launch sites and reusable rockets
ISRO and India's Space Missions	ISRO, major centres, PSLV, GSLV, Chandrayaan, Mangalyaan, Aditya-L1 and Gaganyaan
Space Exploration	astronauts, space stations, Moon and Mars missions, space agencies and famous missions
Basic Physics for Space Science	force, motion, gravity, Newton's Laws, work, energy, heat, light, sound, electricity and magnetism
Scientific Reasoning and Logical Aptitude	observation, patterns, data interpretation, analytical thinking and numerical aptitude
General Space Awareness	current missions, discoveries, scientists, astronauts, events and applications of space technology

Instructions for Students

Choose the best answer from A, B, C and D.

Use rough work for calculations.

Read data and sequence questions carefully.

Do not spend too long on one question; return to it later.

Attempt all ten Olympiad Challenge questions.

Information Cut-off

Mission-status statements were checked against official ISRO and NASA information available through 23 July 2026. Comp testing and future missions are described accordingly.

MOCK PAPER 1

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes Maximum Marks: 50

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for e

1. Astronomy is the study of

- A. objects and events in space B. plants in a garden C. only rocks under Earth D. languages spoken by people

2. Which object is usually easiest to see in the daytime sky?

- A. the Milky Way B. the Sun C. a distant galaxy D. a faint constellation

3. A group of stars that seems to form a pattern is called a

- A. satellite B. meteorite C. constellation D. launch pad

4. Which instrument helps us see faint and distant objects in the sky?

- A. microscope B. thermometer C. compass only D. telescope

5. The Solar System formed from a large cloud of gas and dust. This is the basic idea of the

- A. solar nebula theory B. shadow theory C. tide theory D. rain cycle

6. The Sun is a

- A. planet B. star C. moon D. comet

7. Which planet is closest to the Sun?

- A. Earth B. Mars C. Mercury D. Jupiter

8. Which of these is a dwarf planet?

- A. Venus B. Saturn C. Earth D. Pluto

9. A space rock that reaches the ground is called a

- A. meteorite B. meteor C. meteoroid only D. comet tail

10. Stars appear bright mainly because they

- A. reflect only moonlight B. give out their own light C. are made of ice D. are close to every observer

11. A large group of stars held together by gravity is a

- A. time zone B. rocket stage C. star cluster D. solar panel

12. A cloud of gas and dust in space is called a

- A. crater B. constellation map C. telescope D. nebula

13. Our Solar System is inside the

- A. Milky Way galaxy B. Andromeda Moon C. Kuiper planet D. Earth cluster

14. Earth's outermost solid layer is the

- A. mantle B. crust C. outer core D. inner core

15. Most weather takes place in the lowest layer of the atmosphere called the

- A. exosphere B. mantle C. troposphere D. magnetosphere

16. Day and night are mainly caused by

- A. Earth's revolution only B. the Moon changing shape C. clouds covering the Sun D. Earth's rotation

17. The Moon looks fully lit from Earth during the

- A. full Moon B. new Moon C. solar eclipse D. first sunrise

18. Which is a natural satellite?

- A. INSAT B. the Moon C. a weather balloon D. a rocket

19. A communication satellite mainly helps to

- A. make oceans salty B. create gravity C. send signals over long distances D. stop Earth rotating

20. A weather satellite helps scientists observe

A. the inside of the Sun only B. underground minerals only C. the sound of stars D. clouds and storms

21. Remote sensing means collecting information about an area

A. from a distance using instruments B. only by touching it C. only by digging D. without using any data

22. A rocket moves upward because hot gases are pushed

A. upward B. downward C. sideways only D. nowhere

23. A launch vehicle is built in stages mainly so that it can

A. carry no fuel B. remain on the launch pad C. drop empty parts and become lighter D. avoid using engines

24. The useful object carried by a rocket is called the

A. shadow B. crater C. constellation D. payload

25. India's major space launch centre at Sriharikota is the

A. Satish Dhawan Space Centre B. Vikram Sarabhai Library C. Aryabhata Observatory on Mars D. Indian Ocean Station

26. ISRO stands for

A. International Satellite Repair Office B. Indian Space Research Organisation C. Indian Science Rocket Orbit D. Interplanetary Space Radi

27. PSLV is best known as an Indian

A. space station B. planet C. launch vehicle D. astronaut suit

28. Chandrayaan missions explore the

A. Sun B. Jupiter C. Milky Way centre D. Moon

29. Mangalyaan was sent to study

A. Mars B. Venus C. Mercury D. Saturn

30. A person trained to travel and work in space is an

A. astronomer only B. astronaut C. architect D. athlete

31. The International Space Station travels around

A. the Sun only B. Mars C. Earth D. the Moon

32. NASA is the space agency of the

A. Moon B. European Union only C. planet Mars D. United States

33. A push or a pull is called a

A. force B. shadow C. orbit D. temperature

34. The force that pulls objects towards Earth is

A. magnetism only B. gravity C. light D. sound

35. Work is done when a force moves an object through a

A. colour B. name C. distance D. temperature only

36. Which object allows light to pass through clearly?

A. wooden board B. brick wall C. metal sheet D. clear glass

37. Complete the pattern: Sun, Moon, Sun, Moon, Sun, __.

A. Moon B. Mars C. rocket D. star cluster

38. Which one does not belong with the others?

A. Mercury B. rocket C. Venus D. Earth

39. Aditya-L1 studies the

A. Moon B. Mars C. Sun D. rings of Saturn only

40. Which daily service commonly uses navigation satellites?

A. boiling water B. growing hair C. making shadows disappear D. finding a route on a phone map

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately more demanding. Read carefully, reason step by step and calc

41. A student faces east at sunrise. Which direction is directly behind the student?

A. west B. north C. south D. up

42. A planet is between Venus and Mars in distance from the Sun. Which planet is it?

A. Mercury B. Earth C. Jupiter D. Saturn

43. A bright object gives out its own light and is much larger than Earth. It is most likely a

A. planet B. moon C. star D. asteroid

44. At 6 p.m. in one place, a city one time zone to the east is usually at about

A. 5 p.m. B. 6 a.m. C. 12 noon D. 7 p.m.

45. A farmer wants a satellite image showing which fields are green. Which technology is most useful?

A. remote sensing B. sound recording C. a launch pad D. a star chart

46. Stage 1 of a rocket has finished burning fuel. Why is it released?

A. to increase air resistance B. to reduce mass for the remaining flight C. to stop all engines D. to pull the rocket back to Earth

47. Which mission-object pair is correctly matched?

A. Mangalyaan - Sun B. Aditya-L1 - Mars C. Chandrayaan - Moon D. Gaganyaan - Jupiter

48. Astronauts in an orbiting space station seem to float mainly because they and the station are

A. outside all gravity B. lighter than air C. pushed up by sunlight D. falling around Earth together

49. A 5 N force moves a toy 4 m in the same direction. The work done is

A. 20 J B. 1.25 J C. 9 J D. 40 J

50. A class records stars seen on four nights: 8, 12, 12 and 15. On which two nights was the number the same?

A. first and second B. second and third C. first and fourth D. third and fourth

MOCK PAPER 2

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes Maximum Marks: 50

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for e

1. The imaginary dome on which sky objects appear to lie is called the

- A. ocean floor B. celestial sphere C. rocket chamber D. planetary crust

2. The direction halfway between north and east is

- A. south-west B. north-west C. north-east D. south-east

3. A notebook used during sky watching should record the time, place and

- A. price of the pencil B. student height only C. colour of the cover D. direction of the object

4. A solar eclipse is an example of an

- A. astronomical event B. ocean current C. soil layer D. simple machine

5. Which planet is famous for its large rings?

- A. Mercury B. Saturn C. Venus D. Mars

6. Which group contains only giant planets?

- A. Mercury, Venus, Earth and Mars B. Earth, Mars, Jupiter and Saturn C. Jupiter, Saturn, Uranus and Neptune D. Venus, Earth, Uranus and

7. The Sun provides Earth mainly with

- A. soil and rocks B. tides only C. oxygen from its surface D. light and heat

8. A comet is mainly made of ice, dust and

- A. rocky material B. liquid metal only C. wood D. clouds from Earth

9. A meteoroid that glows while moving through the atmosphere is seen as a

- A. planet B. meteor C. moon D. galaxy

10. A star can look brighter than another because it may be larger, hotter or

- A. made of soil B. inside the Moon C. closer to Earth D. moving slower than sound

11. A constellation is a pattern of stars seen from

- A. inside the Sun B. under the ocean C. inside a rocket engine D. Earth

12. The Milky Way contains

- A. billions of stars B. only eight planets C. one single star D. no gas or dust

13. The Universe includes

- A. only Earth and Moon B. galaxies, stars, planets and space C. only the Solar System D. only objects we can touch

14. Earth's thick layer below the crust is the

- A. troposphere B. inner core only C. mantle D. ocean

15. Earth takes about one year to complete one

- A. rotation on its axis B. Moon phase C. solar eclipse D. revolution around the Sun

16. Seasons occur mainly because Earth is tilted and

- A. revolves around the Sun B. stops rotating C. moves closer to the Moon every month D. changes its mass

17. During a lunar eclipse, which object is between the Sun and Moon?

- A. Mercury B. Earth C. a comet D. the space station

18. A navigation satellite helps a traveller find

- A. the age of a star B. the taste of water C. location and direction D. the weight of sunlight

19. A satellite used for television programmes is mainly a

- A. natural satellite B. meteorological balloon only C. launch vehicle D. communication satellite

20. Satellite images can help during floods by showing

A. which areas are covered by water B. the sound of rain C. the inside of every house D. the names of all people

21. Solar panels on a satellite change sunlight into

A. sound energy only B. electrical energy C. soil D. gravity

22. The place from which a rocket is sent into space is a

A. galaxy B. crater C. launch site D. constellation

23. Rocket fuel and an oxidiser react in the engine to produce hot

A. ice B. sand C. moonlight D. gases

24. A reusable rocket is designed so that some part can

A. fly again after recovery B. turn into a satellite C. remain in space forever D. avoid all maintenance

25. The part at the top of a launch vehicle may protect the payload with a

A. crater B. fairing C. compass D. tide

26. The Vikram Sarabhai Space Centre is closely connected with

A. ocean fishing B. weather on Jupiter only C. launch vehicle development D. building railway engines

27. GSLV is designed to launch satellites, including some into

A. underground tunnels B. the ocean floor C. inside the Sun D. high Earth orbits

28. Aditya-L1 observes the Sun from near the Sun-Earth

A. L1 point B. north pole C. asteroid belt D. Moon crater

29. Gaganyaan is India's programme for

A. underwater farming B. human spaceflight C. building a Moon-sized telescope D. moving Earth

30. A space station is a place where astronauts can

A. stand on the Sun B. stop all satellites C. live and carry out experiments in orbit D. create planets

31. The European Space Agency is commonly shortened to

A. ESO only B. EGA C. EVA D. ESA

32. A rover is a vehicle designed to

A. move across the surface of another world B. orbit only without landing C. measure sound in a classroom D. launch itself without a rocket

33. Newton's first law is also called the law of

A. reflection B. inertia C. tides D. heat

34. When the same force acts on a lighter object, it usually produces

A. less motion in every case B. no change at all C. greater acceleration D. a lower temperature only

35. Sound needs a material medium such as air, but light can travel through

A. only water B. only metal C. nothing at all D. vacuum

36. A magnet has two ends called

A. poles B. orbits C. stages D. phases

37. Which sequence is in increasing order?

A. 8, 6, 4, 2 B. 2, 4, 6, 8 C. 2, 6, 4, 8 D. 4, 2, 8, 6

38. A chart shows 3 rockets on Monday, 5 on Tuesday and 4 on Wednesday. The highest number is on

A. Monday B. Wednesday C. Tuesday D. all three days

39. Chandrayaan-3 demonstrated a safe soft landing on the

A. Sun B. Mars C. Venus D. Moon

40. Which technology helps television channels send programmes across large regions?

A. communication satellites B. meteorites C. constellations D. tides

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately more demanding. Read carefully, reason step by step and calc

41. A star is seen low in the western sky after sunset. Which observation is most complete?

A. only the star colour B. time, place, western direction and height above horizon C. only the observer name D. only the date

42. A body is too small to be a planet, orbits the Sun beyond Neptune, and is round. It is most likely a

A. natural satellite of Earth B. star cluster C. dwarf planet D. rocket stage

43. Three objects are listed: a nebula, a star cluster and a galaxy. Which is largest in general?

A. nebula always B. star cluster always C. all must be exactly equal D. galaxy

44. It is summer in the Northern Hemisphere. Which statement is most likely true?

A. the Northern Hemisphere is tilted more towards the Sun B. Earth is much closer to the Sun only C. the Moon blocks winter D. Earth has s

45. A satellite must photograph the same farming region every few days. Which ability is most important?

A. producing rocket thrust B. repeated Earth observation from orbit C. making a comet tail D. changing time zones

46. A two-stage rocket has used all fuel in stage one. Keeping the empty stage attached would mainly

A. create extra useful fuel B. remove gravity C. make later acceleration harder D. increase payload mass without cost

47. Which statement correctly compares PSLV and GSLV?

A. PSLV is a planet and GSLV is a star B. both are space stations C. GSLV studies only oceans D. both are Indian launch vehicles used for

48. Why do astronauts exercise regularly on a space station?

A. to reduce loss of muscle and bone strength B. to create gravity for Earth C. to recharge the Sun D. to move the station by running

49. A 12 N force moves a box 3 m. How much work is done when force and motion are in the same direction?

A. 4 J B. 36 J C. 15 J D. 144 J

50. A pattern follows: 3, 6, 12, 24, __. What is next?

A. 27 B. 30 C. 48 D. 36

MOCK PAPER 3

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes Maximum Marks: 50

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for e

1. At night, stars appear to move from east to west mainly because

- A. stars circle Earth once each minute B. the Sun pushes stars C. Earth rotates from west to east D. the Moon turns the sky

2. A compass is useful in sky observation because it helps find

- A. star temperatures directly B. planet sizes C. rocket fuel D. directions

3. Which event can be safely watched only with proper solar protection?

- A. a solar eclipse B. a full Moon C. a meteor shower D. a constellation

4. A telescope with a larger opening usually collects

- A. less light B. more light C. only sound D. no information

5. Earth is the ___ planet from the Sun.

- A. first B. fifth C. third D. eighth

6. Mars is often called the Red Planet because of iron-rich material in its

- A. clouds of water only B. rings C. comet tail D. surface rocks and dust

7. Which planet is the largest?

- A. Jupiter B. Earth C. Mercury D. Mars

8. Most asteroids in our Solar System orbit between

- A. Earth and Moon B. Mars and Jupiter C. Sun and Mercury D. Uranus and Neptune only

9. A comet develops a visible tail when it comes closer to the

- A. Earth only B. Milky Way centre C. Sun D. north pole

10. Stars differ in colour partly because they have different

- A. numbers of moons B. time zones C. rocket stages D. surface temperatures

11. A blue-white star is generally hotter than a

- A. red star B. blue star C. white star D. star with no colour

12. Many young stars can form inside a

- A. meteorite B. nebula C. satellite dish D. launch pad

13. A galaxy is held together mainly by

- A. sound B. air pressure C. gravity D. sunlight only

14. Earth's centre contains the

- A. crust only B. atmosphere C. ocean floor only D. core

15. The ozone layer helps absorb much of the Sun's harmful

- A. ultraviolet radiation B. radio sound C. gravity D. visible green light only

16. Earth completes one rotation in about

- A. 7 days B. 24 hours C. 30 days D. 365 days

17. The Moon phase that comes after new Moon begins as a

- A. waning crescent B. full Moon C. waxing crescent D. lunar eclipse

18. A satellite orbiting Earth follows a curved path because of its motion and

- A. sound waves B. clouds C. magnetic paint D. gravity

19. Navigation satellites are used by ships, aircraft and phones to find

- A. position B. star colour C. rocket temperature only D. Moon age

20. Remote sensing can help map

A. only constellations B. forests, farms and water bodies C. only rocket engines D. only sounds

21. A spacecraft needs antennas mainly to

A. make craters B. produce tides C. communicate with Earth D. form stars

22. The upward push that moves a rocket is called

A. tide B. shadow C. orbit D. thrust

23. The first rocket stage usually operates

A. soon after lift-off B. after the payload lands C. only on the Moon D. before fuel is added

24. A payload fairing protects a satellite mainly during

A. its entire life under the ocean B. flight through the atmosphere C. a lunar eclipse D. a star forming

25. A reusable launch system can lower costs by

A. making every rocket heavier B. removing all testing C. using recovered parts again D. avoiding navigation

26. The main headquarters of ISRO is in

A. New York B. Tokyo C. the Moon D. Bengaluru

27. PSLV launched India's Mars Orbiter Mission towards

A. Mars B. Jupiter C. the Sun D. Neptune

28. Aditya-L1 is India's first dedicated space mission to observe the

A. Moon B. Sun C. Earth's oceans only D. asteroid belt

29. Gaganyaan spacecraft systems are being developed to carry

A. only weather balloons B. comets C. Indian astronauts D. ocean ships

30. A space suit provides an astronaut with air and

A. a natural atmosphere B. a Moon crater C. a new planet D. protection from the space environment

31. Mars rovers are controlled using signals sent from

A. Earth B. the Sun's core C. inside Jupiter D. the Milky Way centre

32. JAXA is the space agency of

A. India B. Japan C. France only D. Brazil only

33. Newton's third law explains that forces come in

A. circles only B. different colours C. equal and opposite pairs D. units of time

34. An object moving faster has more kinetic energy if its mass is

A. zero B. unknown only C. always smaller D. the same

35. Heat naturally flows from a hotter object to a

A. colder object B. hotter object only C. star only D. vacuum only

36. An electric circuit needs a complete path for

A. gravity to stop B. current to flow C. light to vanish D. sound to freeze

37. Which number should replace the blank? 5, 10, 15, __, 25

A. 18 B. 30 C. 20 D. 35

38. A table shows temperatures of 20°C, 25°C, 22°C and 25° C. The highest value occurs

A. once B. three times C. four times D. twice

39. The Chandrayaan-3 landing place was in the Moon's

A. southern high-latitude region B. equatorial ocean C. atmosphere D. north pole of Earth

40. A modern application of satellites is

A. making stars B. tracking cyclones C. stopping seasons D. turning planets into moons

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately more demanding. Read carefully, reason step by step and calc

41. A star is 20° above the northern horizon. Which pair describes it correctly?

A. altitude north and direction 20° B. altitude 70° and direction south C. altitude 20° and direction north D. distance 20 km and direction north

42. Which order is correct from the Sun outward?

A. Venus, Mercury, Mars, Earth B. Earth, Venus, Mercury, Mars C. Mercury, Earth, Venus, Mars D. Mercury, Venus, Earth, Mars

43. Object A is a cloud of gas and dust. Object B is a huge group of stars. A and B are respectively

A. nebula and galaxy B. galaxy and constellation C. star cluster and planet D. meteor and nebula

44. If it is 12 noon at longitude 0°, the basic time-zone rule gives about 2 p.m. at 30° east. Why?

A. Earth stops at noon B. Earth rotates about 15° per hour C. the Moon adds two hours D. east always means midnight

45. A satellite image shows less green area this year than last year. The most reasonable first conclusion is that

A. gravity has stopped B. the satellite made the land brown C. vegetation cover may have decreased D. all plants on Earth are gone

46. A rocket produces 800 units of thrust and has 300 units of opposing force. The unbalanced upward force is

A. 1100 units B. 300 units C. 240000 units D. 500 units

47. Which statement is accurate in 2026?

A. Gaganyaan is undergoing development and qualification tests B. Gaganyaan has already landed people on Mars C. Aditya-L1 studies Jupiter

48. A message from Earth takes several minutes to reach a Mars rover. This means the rover must

A. wait for a person beside it B. perform some actions automatically C. travel faster than light D. use sound through space

49. A 2 kg object moves at 3 m/s. Using momentum = mass x speed, its momentum is

A. 1.5 kg m/s B. 5 kg m/s C. 6 kg m/s D. 9 kg m/s

50. A student changes both the lamp distance and screen material while testing shadow size. Why is the test weak?

A. the lamp gives light B. shadows can be measured C. the screen has a surface D. more than one variable was changed

MOCK PAPER 4

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes Maximum Marks: 50

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for e

1. The line where the sky seems to meet the ground is the

- A. zenith B. orbit C. equator only D. horizon

2. The point directly overhead in the sky is called the

- A. zenith B. nadir C. horizon D. eclipse

3. Which is a good condition for observing faint stars?

- A. a brightly lit stadium B. a dark place away from city lights C. looking through thick clouds D. standing under a streetlamp

4. A meteor shower is caused when Earth passes through

- A. a star cluster B. the Sun's surface C. a stream of small particles D. a rocket launch pad

5. Which planet has the hottest average surface because of its thick atmosphere?

- A. Mercury B. Mars C. Neptune D. Venus

6. Uranus and Neptune are often called

- A. ice giants B. terrestrial planets C. dwarf stars D. natural satellites

7. A natural satellite moves around a

- A. constellation B. planet or dwarf planet C. galaxy only D. rocket stage

8. The asteroid belt is a region containing many rocky bodies orbiting the

- A. Moon B. Earth only C. Sun D. Milky Way centre

9. A meteoroid is found mainly

- A. only on Earth's ground B. inside a telescope C. inside a star D. in space before entering an atmosphere

10. Which property tells how hot a star's surface is most directly?

- A. colour B. number of planets only C. constellation name D. distance from school

11. A group of stars born from the same cloud may form a

- A. time zone B. star cluster C. launch fairing D. meteorite field only

12. The Milky Way appears as a faint band because we are viewing many distant

- A. oceans B. rockets C. stars D. Moon phases

13. Andromeda is another

- A. planet B. constellation only C. rocket engine D. galaxy

14. The layer of gases around Earth is the

- A. atmosphere B. crust C. mantle D. core

15. When the Northern Hemisphere is tilted away from the Sun, it receives

- A. more direct sunlight B. less direct sunlight C. no sunlight anywhere D. light only from the Moon

16. A first-quarter Moon appears approximately

- A. fully dark B. fully lit C. half lit D. as a solar eclipse

17. Ocean tides are caused mainly by the gravity of the

- A. North Star B. asteroid belt C. International Space Station D. Moon

18. A weather satellite usually carries instruments that observe clouds and

- A. temperature patterns B. star birth only C. rocket paint D. Moon rocks only

19. A navigation receiver uses signals from several satellites to calculate its

- A. mass B. position C. colour D. temperature of the Sun

20. An Earth-observation satellite can help map damage after a

A. constellation change B. solar-system formation C. cyclone D. star cluster birth

21. The main body of a spacecraft that holds systems and instruments is called the

A. meteor train B. launch shadow C. planet crust D. spacecraft bus

22. The engine nozzle directs hot gases mainly

A. backward B. towards the payload C. into the fuel tank D. straight into the Sun

23. A rocket stage includes engines, fuel tanks and

A. oceans B. supporting structures C. constellations D. Moon phases

24. The payload is separated from the launch vehicle after reaching the required

A. classroom B. continent C. path or orbit D. star colour

25. A reusable booster may return by landing or

A. becoming a comet B. entering the Sun C. turning into a planet D. splashing down for recovery

26. Vikram Sarabhai is widely known as a key founder of India's

A. space programme B. ocean programme C. railway network D. weather seasons

27. GSLV uses powerful stages to place satellites into

A. the ocean floor B. higher-energy orbits C. Moon craters without launch D. inside Earth's core

28. Chandrayaan-2 included an orbiter designed to study the

A. Sun B. Mars C. Moon D. Saturn

29. The aim of Gaganyaan includes sending Indian crew to

A. the surface of the Sun B. another galaxy C. inside Jupiter D. low Earth orbit

30. Astronauts train in water partly because it can help practise

A. movement during spacewalk-like tasks B. planet formation C. solar eclipses D. building stars

31. A Moon mission may use an orbiter, lander and

A. tide B. rover C. constellation D. time zone

32. Roscosmos is the space agency of

A. India B. Japan C. Russia D. Canada only

33. A moving object remains in motion unless an unbalanced force acts. This describes

A. Newton's third law only B. reflection C. conduction D. Newton's first law

34. Potential energy is energy stored because of position or

A. condition B. colour only C. name D. sound level only

35. A mirror changes the direction of light by

A. conduction B. reflection C. evaporation D. gravity

36. An electromagnet works when electric current flows through a

A. wooden block B. glass of water only C. coil of wire D. shadow

37. Which fraction is largest?

A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{2}{5}$ D. $\frac{3}{4}$

38. A student measures a shadow at 9 a.m., noon and 3 p.m. The shadow is usually shortest near

A. noon B. 9 a.m. only C. 3 p.m. only D. midnight

39. Aditya-L1 scientific data help researchers study solar activity and

A. ocean tides only B. space weather C. Earth's core D. meteorite names

40. One benefit of space technology during disasters is

A. stopping all storms B. removing gravity C. rapid communication and mapping D. changing Earth's orbit

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately more demanding. Read carefully, reason step by step and calc

41. A student sees a star exactly halfway between the horizon and the zenith. Its altitude is about

A. 0° B. 90° C. 180° D. 45°

42. A small body orbits the Sun, is mostly ice and forms a tail near the Sun. It is a

A. comet B. asteroid only C. moon D. dwarf star

43. Which order goes from smaller system to larger system?

A. Universe, galaxy, star cluster B. star cluster, galaxy, Universe C. galaxy, Universe, star cluster D. star cluster, Universe, galaxy

44. A place at 45° east is about three hours ahead of 0° longitude because

A. the Moon moves 15° each hour B. seasons add hours C. Earth rotates about 15° each hour D. eclipses move clocks

45. Satellite A stays above nearly the same region; Satellite B passes over many latitudes. A and B are most likely

A. both natural satellites B. both fixed on the ground C. polar-orbiting and launch vehicle D. geostationary-type and polar-orbiting

46. A rocket has three stages. Which stage must usually fire first?

A. the lowest stage at lift-off B. the payload C. the top stage after landing D. all stages only after orbit

47. Which mission status is correctly stated?

A. Chandrayaan-3 is orbiting Mars B. Aditya-L1 is operating as a solar observatory C. Gaganyaan has landed humans on the Moon D. PSLV

48. A rover receives a command 10 minutes after it is sent. An immediate hazard appears before the command arrives. The rover n

A. a louder speaker B. a shorter planet C. onboard hazard-detection software D. sound waves through vacuum

49. A 4 kg object speeds up from 1 m/s to 3 m/s. Its momentum increases by

A. 2 kg m/s B. 4 kg m/s C. 12 kg m/s D. 8 kg m/s

50. A plant experiment uses identical pots but different amounts of water. What should be kept the same?

A. plant type, soil, light and time B. water amount C. final height only D. result

MOCK PAPER 5

ISO 2026 - Junior Level (Classes V-VI)

Time: 60 minutes Maximum Marks: 50

Questions 1-40: Core Section. Questions 41-50: Olympiad Challenge. Select one option for e

1. North, south, east and west are called the four main

A. directions B. phases C. orbits D. galaxies

2. A star chart is used to help identify

A. rocket fuel B. stars and constellations C. Earth's core D. satellite batteries

3. Binoculars can help beginners observe the Moon because they

A. create moonlight B. stop clouds C. magnify distant objects D. change gravity

4. The best reason to repeat a sky observation on several nights is to

A. make the sky move faster B. increase the number of planets C. change the weather D. notice patterns and changes

5. Which planet is known for having a Great Red Spot?

A. Jupiter B. Venus C. Mercury D. Earth

6. Which planet is farthest from the Sun among the eight planets?

A. Mars B. Neptune C. Saturn D. Uranus

7. Earth's Moon is a

A. dwarf planet B. star C. natural satellite D. comet

8. Ceres is found mainly in the

A. Sun's atmosphere B. Kuiper Belt beyond Neptune only C. Earth's ocean D. asteroid belt

9. A meteor is commonly called a

A. shooting star B. new planet C. space station D. Moon phase

10. A red star is usually cooler than a

A. red star of the same type B. blue star C. planet D. meteorite

11. Stars in a constellation may look close together but can actually be at very different

A. temperatures only B. time zones C. distances D. rocket stages

12. The Milky Way is only one of many

A. moons B. planets C. launch sites D. galaxies

13. The space between galaxies is part of the

A. Universe B. Earth's crust C. rocket fairing D. troposphere only

14. The inner core of Earth is mainly solid because of very high

A. rainfall B. pressure C. moonlight D. wind speed

15. Earth's atmosphere contains the air needed by

A. stars B. asteroids C. living things D. rocket stages only

16. The Moon takes about one month to go around

A. the Sun directly only B. Mars C. Jupiter D. Earth

17. A solar eclipse happens when the Moon blocks part or all of the

A. Sun B. Earth C. Milky Way D. north direction

18. A satellite that studies crops and forests is an

A. natural satellite B. Earth-observation satellite C. astronaut D. launch pad

19. A communication satellite often receives, strengthens and sends

A. gravity B. tides C. signals D. meteorites

20. Weather forecasts improve when satellites provide pictures of

A. star clusters B. rocket stages C. Moon rocks only D. cloud movement

21. A satellite needs a power system, communication system and

A. control system B. ocean system C. constellation system D. tide system

22. A rocket must carry its own oxidiser because space has almost no

A. gravity B. air C. light D. temperature

23. The first few seconds after launch require strong thrust to overcome gravity and

A. Moon phases B. time zones C. air resistance D. star colour

24. The final rocket stage may place a satellite into the correct

A. continent B. ocean C. constellation D. orbit

25. A payload can be a satellite, spacecraft or scientific

A. instrument B. season C. shadow only D. time zone

26. Aryabhata was India's first

A. crew member B. satellite C. Moon rover D. launch site

27. The Chandrayaan-3 rover was named

A. Vikram B. Rohini C. Pragyan D. Bhaskara

28. The Chandrayaan-3 lander was named

A. Pragyan B. Mangalyaan C. Aryabhata D. Vikram

29. Aditya-L1 was launched using a

A. PSLV B. submarine C. weather balloon D. Moon rover

30. An astronaut outside a spacecraft wears a suit that supplies

A. soil B. oxygen C. rain D. natural gravity

31. Space probes can visit worlds that are too distant for

A. radio signals B. light C. human crews at present D. gravity

32. The China National Space Administration is commonly shortened to

A. ISRO B. NASA C. JAXA D. CNSA

33. Friction is a force that usually opposes

A. motion B. mass C. time D. temperature

34. Energy stored in a stretched rubber band is

A. sound only B. potential energy C. light only D. nuclear energy only

35. White light can split into colours when it passes through a

A. metal block B. magnet C. prism D. battery only

36. Two like poles of magnets

A. attract each other B. produce gravity only C. become light D. repel each other

37. Which shape comes next: circle, square, circle, square, ___?

A. circle B. triangle C. rectangle D. star only

38. A graph rises from 2 to 5 and then to 8. The increase each step is

A. 2 B. 3 C. 5 D. 8

39. In 2026, Gaganyaan was still undergoing important

A. crewed landings on Mars B. construction on the Sun C. tests and qualification work D. flights to another galaxy

40. Satellites help fishermen and sailors through weather information and

A. Moon phases only B. star formation C. rocket thrust D. navigation

OLYMPIAD CHALLENGE - QUESTIONS 41-50

These ten questions are deliberately more demanding. Read carefully, reason step by step and calc

41. An observer turns from north through a quarter turn clockwise. The observer now faces

A. east B. west C. south D. north-west

42. Planet A is rocky and close to the Sun. Planet B is very large and has many moons. A and B are most likely

A. two dwarf planets B. terrestrial planet and giant planet C. two stars D. comet and meteor

43. If a star looks faint, which conclusion is safest?

A. it must be the smallest star B. it must be outside the Universe C. it may be far away, small, cool, or a combination D. it gives no light

44. The Moon is full tonight. About one week later it will be near the

A. new Moon immediately B. first-quarter phase C. solar eclipse every time D. third-quarter phase

45. A phone finds its position using signals from four navigation satellites. Why are several satellites useful?

A. they allow the receiver to compare signal travel information B. one satellite creates all gravity C. they make the phone heavier D. they stop the phone from moving

46. A rocket's total upward thrust is 1000 N while gravity and drag together act downward with 700 N. The net force is

A. 1700 N upward B. 300 N upward C. 300 N downward D. 700 N upward

47. Which pair is correctly matched?

A. Vikram - Mars Orbiter B. Aditya-L1 - lunar rover C. Pragyan - Chandrayaan-3 rover D. PSLV - space station

48. Why can a spacecraft not use ordinary sound to talk directly through empty space?

A. sound is too bright B. space has too much water C. gravity blocks all sound equipment D. sound needs a material medium

49. A battery lights two identical bulbs in series. If one bulb is removed and the circuit breaks, the other bulb goes out because

A. the path for current is incomplete B. gravity is removed C. light cannot reflect D. the battery becomes a magnet only

50. A student records rocket-model distances of 6 m, 8 m, 8 m, 9 m and 11 m. The mode is

A. 6 m B. 8 m C. 9 m D. 11 m

ANSWER KEYS

Mock Paper 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A

26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B

Mock Paper 2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B

26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C

Mock Paper 3

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C

26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D

Mock Paper 4

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D

26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A

Mock Paper 5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A

26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D	A	B

OLYMPIAD CHALLENGE - ANSWER EXPLANAT

Mock Paper 1

41. **A** - East and west are opposite directions, so west is directly behind a person facing east.
42. **B** - The planetary order begins Mercury, Venus, Earth, Mars.
43. **C** - Stars produce their own light, while planets and moons mainly reflect light.
44. **D** - Moving one time zone east usually adds one hour.
45. **A** - Remote sensing instruments collect information about Earth from a distance.
46. **B** - Dropping an empty stage reduces the mass that later stages must accelerate.
47. **C** - Chandrayaan is India's lunar programme.
48. **D** - The station and astronauts continuously fall around Earth in orbit, producing apparent weightlessness.
49. **A** - Work = force x distance = $5 \times 4 = 20$ joules.
50. **B** - The repeated value is 12, recorded on the second and third nights.

Mock Paper 2

41. **B** - A useful observation records when and where the object was seen and its position in the sky.
42. **C** - A round body orbiting the Sun that has not cleared its orbital region may be classified as a dwarf planet.
43. **D** - A galaxy can contain billions of stars along with clusters and nebulae.
44. **A** - Seasonal heating depends mainly on Earth's axial tilt and the angle of sunlight.
45. **B** - Earth-observation satellites revisit regions and collect comparable images.
46. **C** - The empty stage adds mass, so later engines must accelerate more unnecessary material.
47. **D** - PSLV and GSLV are launch-vehicle families designed for different payloads and orbits.
48. **A** - Long stays in microgravity can weaken muscles and bones, so exercise is essential.
49. **B** - Work = force x distance = $12 \times 3 = 36$ joules.
50. **C** - Each number is twice the previous number.

Mock Paper 3

41. **C** - Altitude is the angle above the horizon; north gives the compass direction.
42. **D** - The first four planets are Mercury, Venus, Earth and Mars.
43. **A** - A nebula is a gas-and-dust cloud; a galaxy is a vast system of stars and other matter.
44. **B** - Earth turns 360° in 24 hours, or about 15° each hour.
45. **C** - A change in green area can indicate vegetation change, but further evidence is needed before a stronger claim.
46. **D** - Net upward force = thrust - opposing force = $800 - 300 = 500$.
47. **A** - ISRO continued Gaganyaan qualification and parachute-related tests in 2026.
48. **B** - Because of communication delay, rovers need onboard software for safe and partly autonomous operation.
49. **C** - Momentum = $2 \times 3 = 6$ kg m/s.
50. **D** - A fair test changes one factor at a time while keeping other conditions the same.

Mock Paper 4

41. **D** - The angle from horizon to zenith is 90° , so halfway is about 45° .
42. **A** - Icy bodies that form tails near the Sun are comets.
43. **B** - A star cluster is within a galaxy, and galaxies are part of the Universe.
44. **C** - 45° divided by 15° per hour gives about 3 hours.
45. **D** - A geostationary satellite appears fixed over one longitude; a polar satellite covers many latitudes.
46. **A** - The lowest or first stage provides initial thrust during lift-off.
47. **B** - Aditya-L1 continues solar observations from near the Sun-Earth L1 point.
48. **C** - Communication delay makes onboard sensing and autonomous safety actions necessary.

49. D - Change in momentum = $m(v_2 - v_1) = 4(3 - 1) = 8 \text{ kg m/s}$.

50. A - Only the amount of water should change; other important conditions should remain controlled.

Mock Paper 5

41. A - A quarter turn clockwise from north points east.

42. B - Rocky inner planets are terrestrial; very large outer planets are giants.

43. C - Apparent brightness depends on several factors, especially luminosity and distance.

44. D - The main phases are about one week apart: full Moon is followed by third quarter.

45. A - Position is found by comparing timing and distance information from multiple satellites.

46. B - Net force = $1000 - 700 = 300 \text{ N}$ upward.

47. C - Pragyan was the Chandrayaan-3 rover; Vikram was the lander.

48. D - Sound waves require matter such as air, while radio waves can travel through vacuum.

49. A - Electric current needs a complete conducting path.

50. B - The mode is the value that occurs most often; 8 m appears twice.

SYLLABUS COVERAGE AUDIT

Unit	Paper 1	Paper 2	Paper 3	Paper 4	Paper 5
Fundamentals of Astronomy	5	5	5	5	5
The Solar System	6	6	6	6	6
Stars, Galaxies and the Universe	5	5	5	5	5
Earth and Moon	5	5	5	5	5
Satellites and Space Technology	5	5	5	5	5
Rockets and Launch Vehicles	5	5	5	5	5
ISRO and India's Space Missions	5	5	5	5	5
Space Exploration	4	4	4	4	4
Basic Physics for Space Science	5	5	5	5	5
Scientific Reasoning and Logical Aptitude	3	3	3	3	3
General Space Awareness	2	2	2	2	2

Difficulty and Scoring Guide

Questions 1-15 are mainly direct recall and recognition. Questions 16-40 combine understanding and application. Questions 41-50 combine step reasoning or short calculations.

Score	Suggested interpretation
45-50	Excellent Olympiad readiness
38-44	Strong preparation; review challenge errors
28-37	Good foundation; revise weaker units
18-27	Developing understanding; practise unit by unit
0-17	Revisit the Junior Level reading material before retesting